

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division

Digital Learning Handout

Part A: Content Design

Course Title	AI and ML for Robotics
Course No(s)	AIML ZG528
Credit Units	4
Credit Model	1.25 - 1.5 - 1.25
Course Author	Raja vadhana Prabhakar
Version No:	3.0
Date:	12/11/2025

Course Description:

Fundamentals of robotics. Aerial robots, warehouse robots, under actuated robots. Sensor systems for robots. Effectors and actuators. Robot Operating System. Robot motion models. PID control, Beam model of range finders. Recursive state estimation - Bayes filters, Kalman, extended Kalman, information filters, and nonparametric filters such as particle filters. Mobile robot localization - extended Kalman filter, Grid and Monte Carlo. Simultaneous mapping and localization algorithms, path-planning algorithms, Instance based learning, demonstration based path planning using reinforcement learning, deep learning and reinforcement learning based mapping, navigation and control of mobile robots.

Course Objectives

No	Course Objective
CO1	Understand the fundamental principles and components of robotics
CO2	Apply probabilistic robotic techniques for localization and mapping
CO3	Understand and apply path planning algorithms for cooperative mobile robots
CO4	Apply machine learning techniques to solve robotic tasks like perception, decision-making and control
CO5	Discuss ethical considerations in human-robot interaction including privacy, safety, and societal impact

Text Book(s):

T1	Dieter Fox, Sebastian Thrun, Wolfram Burgard, “Probabilistic Robotics”, Intelligent Robotics and Autonomous Agents series Ed, MIT Press, 2005
T2	Bruno Siciliano, Giuseppe Oriolo, Lorenzo Sciavicco, Luigi Villani, “Robotics: Modelling, Planning and Control”, First Ed, Springer London Ltd, 2009

Reference Book(s) & other resources:

R1	Stuart Russell and Peter Norvig, “Artificial Intelligence – A Modern Approach”, Third Ed, Pearson Education, 2010
R2	Ester Martinez Martin, “Swarm Robotics from Biology to Robotics”, First Ed, IntechOpen, 2010
R3	Ian Goodfellow, Yoshua Bengio, Aaron Courville, “Deep Learning”, MIT Press, 2017





R4	Patrick Lin, Keith Abney, Ryan Jenkins, “Robot Ethics 2.0: From Autonomous Cars to Artificial Intelligence”, Second Ed , Oxford University Press, 2017
R5	Ramkumar Gandhinathan, “ROS Robotics Projects”, Second Ed, Packt Publishing, 2019
R6	Francis X. Govers III, Dr. Kamesh Namuduri, “Artificial Intelligence for Robotics”, Second Ed, Packt Publishing, 2024

Learning Outcomes: Students will be able to

LO1	Identify the necessary components of robotic systems to focus for specific software solution
LO2	Implement Bayesian filters for recursive state estimation and Simultaneous Localization And Mapping algorithms for mapping and localization in mobile robotics environments
LO3	Compare traditional motion planning search algorithms like A* variants , Rapidly exploring Random Tree, Probabilistic Road Maps and implement for mobile robots using python programming
LO4	Implement artificial intelligence and machine learning techniques like swarm intelligence, Convolution Neural Networks, Deep Reinforcement Learning for path planning, mapping and robot control
LO5	Critically assess the applicability, strengths and limitations of current machine learning methodologies, including their applicability to different types of robot sensor data based analytics
LO6	Identify ethical issues in the application of AI and ML techniques in robotics for human interaction

Modular Content Structure

1. Introduction:
 - Evolution of Robotics - Types of Robots: Aerial, Warehouse, Under Actuated, Mobile - Applications and challenges.
2. Components of Programmable Robotic System:
 - Robot Motion: Probabilistic Kinematics – Motion Models
 - Robot Perception: Actuators – Sensors – Types - Proprioceptive - Exteroceptive
 - Maps: Representations - Feature based vs Location based, Range Sensors – Beam Models – Map Matching – Measurement models: Correlation based vs Feature Based models
 - Control Architecture: Functional Architecture – Programming Environment
3. Localization:
 - Taxonomy of Localization Problems
 - Markov Localization
 - Grid Localization – Monte Carlo Localization
 - Localization Techniques: GPS based, Odometry based, Sensor Fusion based
 - Machine Learning for Robot Localization
4. Mapping:
 - Introduction – Occupancy Grid Mapping
 - Simultaneous Localization And Mapping (SLAM) - Feature-based vs Grid-based SLAM – Graph SLAM – Fast SLAM
 - Mobile Robot Visual SLAM Based on Neural Network





5. Path Planning:

- Introduction, Graph based vs Sampling based algorithms
- Graph based path planner: A* - D*
- Sampling based path planner: Rapidly Exploring Random Trees(RRT) – RRT* - Probabilistic Road Maps (PRM)
- Swarm Intelligence: Cobots – Ant Colony Optimization – Particle Swarm Optimization – Genetic Algorithm – Firefly Algorithm
- Deep Reinforcement Learning for Path Planning

6. Human – Robot Interaction:

- Introduction – Modalities – Applications – Ethical considerations
- Visual bot: Hand/Body detection – Pose estimation – Gesture recognition – Static vs Dynamic - Gesture Classification Agent
- Chat bot: Large Language Models – Retrieval Augmented Generation – Conversational Agent

Part B: Learning Plan

Contact Session	List of Topic Title	Sub-Topics	Reference
1	Introduction	● Introduction to Robotics, Type of Robots – Mobile – Industrial – Field - Service ● Brief Overview of Robot Modelling, Planning, Control ● Application and Challenges : Probabilistic Robotics	T2: Chapter 1 T1: 1.1 - 1.3
2	Components of Programmable Robotic System	● Robot Motion: Overview of Probabilistic Motion Models – Motion and Maps ● Robot Perception: Actuators – Sensors – Types - Proprioceptive - Exteroceptive	T1: Chapter 5 T2: Chapter 5
3	Components of Programmable Robotic System	● Maps – Representations - Feature based vs Location based ● Beam Models for Range Finders ● Likelihood Fields - Map Matching	T1: Chapter 6





		Measurement models: Correlation based vs Feature Based models	
4	Components of Programmable Robotic System	- Control Architecture - Functional Architecture - Programming Environment	T2: Chapter 6.1, 6.2
5	Localization	- Mobile Robot Localization - Taxonomy of Localization Problems - Markov Localization	T1: Chapter 7
6	Localization	- Grid Localization - Monte Carlo Localization - Machine Learning for Robot Localization Using Fiducial Markers	T1: Chapter 8, Research papers & web resources https://www.mdpi.com/1424-8220/23/6/3128
7	Localization	- Mobile Robot Localization - Localization Techniques: GPS based, Odometry based, Sensor Fusion based - Deep Learning for Odometric Learning - Deep Learning-Based Multi-Sensor Fusion Techniques	Research papers & web resources Novel lightweight odometric learning method for humanoid robot localization - ScienceDirect
8	Mapping	- Mapping: Introduction - Occupancy Grid Mapping - Simultaneous Localization And Mapping (SLAM) - Feature-based vs Grid-based SLAM	T1: Chapter 9, 10, 11, 13
9	Mapping	- Graph SLAM - Fast SLAM - Mobile Robot Visual SLAM Based on Neural Network	T1: Chapter 9, 10, 11, 13 Research papers & web resources: https://www.frontiersin.org/journals/robotics-and-





			ai/articles/10.3389/frobt.2024.1347985/full
10	Path Planning	• Path Planning • Introduction, Graph based vs Sampling based algorithms • Graph based path planner: A* - D*	T2: Chapter 12 Research papers & web resources: Electronics and Robotics Club (ERC), BITS Goa - Handbook (erc-bpgc.github.io)
11	Path Planning	• Sampling based path planner • Rapidly Exploring Random Trees(RRT) – RRT* • Probabilistic Road Maps (PRM)	T2: Chapter 12 Research papers & web resources: Electronics and Robotics Club (ERC), BITS Goa - Handbook (erc-bpgc.github.io)
12	Path Planning	• Swarm Intelligence: Cobots • Ant Colony Optimization – Particle Swarm Optimization	R2: Chapters
13	Path Planning	• Genetic Algorithm • Firefly Algorithm	Research papers & web resources: https://doi.org/10.1016/j.dt.2018.06.004
14	Human – Robot Interaction	• Human – Robot Interaction • Introduction – Modalities - Applications – Ethical Considerations • Visual bot: Hand/Body detection – Pose estimation – Gesture recognition – Static vs Dynamic - Gesture Classification Agent	Research papers & Class Materials: https://doi.org/10.1109/THMS.2022.3227309
15	Human – Robot Interaction	• Visual bot: Gesture recognition – Static vs Dynamic - Gesture Classification Agent • Chat bot: Large Language	Research papers & Class Materials: https://doi.org/10.1109/RICAI60863.2023.10489289 https://doi.org/10.1109/SSD





		Models – Retrieval Augmented Generation – Conversational Agent	61670.2024.10548649
16	Ethics in AI	Review	

Experiential Learning Components:

1. Lab work: 7
2. Project work: 0
3. Case Study: 0
4. Simulation: 0
5. Work Integrated Learning Assignment- 2 Assignments
6. Design work/ Field work: 0

Objective of Experiential Learning Component:

Learners will implement machine learning or deep learning techniques for software mobile robots in simulated environment to perform tasks like localization, mapping and path planning.

Scope of Experiential Learning Component:

Programming language – Python

Tools and libraries: PyTorch, TensorFlow (with Keras API), Stable Baselines3 (SB3), Scikit-learn, OpenCV , NumPy, SciPy

Lab Infrastructure:

** Included only open source alternatives*

Framework: (Any one of) GoogleColab/ ROS2/ Autoware/ PythonRobotics

Simulation: (Any one of) PyBullet/ MuJoCo, OpenAI Gymnasium/Gazebo/ RViz

List of Experiments:

Lab No	Lab Objective	Session Reference
1	Install and Getting Started with robotic framework	4
2	Understand the necessary components of a robotics system & set up simulation environment for coding	4
3	Implement AIML models in Sensor & Motion Modelling	2,3,4
4	Implement Machine Learning for Robot Localization	6
5	Implement Visual SLAM using Convolution Neural Network	9
6	Implement sampling based planner and evolutionary techniques for Path Planning	12
7	Implement Deep Reinforcement Learning for Path Planning	14





Evaluation Scheme:

Legend: EC = Evaluation Component; AN = After Noon Session; FN = Fore Noon Session

Evaluation Component	Name (Quiz, Lab, Project, Mid-term exam, End semester exam, etc.)	Type (Open book, Closed book, Online, etc.)	Weight	Duration	Day, Date, Session, Time
EC – 1*	Quiz I	Online	5%	1 day	To be announced
	Assignment/Lab Assignment	Online	10% - 15%	10 days	To be announced
	Assignment/Lab Assignment & Situated Learning	Online	15 % - 20%	10 days	To be announced
EC - 2	Mid-Semester Test	Closed Book	30%	2 hours	To be announced
EC - 3	Comprehensive Exam	Open Book	35% - 40%	2 ½ Hours	To be announced

EC1* (30-35%): Quiz: 5 %, Lab Assignment/Assignment: 20-25%

Syllabus for Mid-Semester Test (Closed Book): Topics in Contact session: 1 to 8

Syllabus for Comprehensive Exam (Open Book): All topics

Important Links and Information:

eLearn Portal: <https://elearn.bits-pilani.ac.in>

Students must visit the eLearn portal regularly and stay updated with the latest announcements and deadlines.

Contact Sessions: Students should attend the online lectures as per the schedule provided on the eLearn portal.

Evaluation Guidelines:

1. EC-1 consists of two Assignments and either one or two Quizzes. Students will attempt them through the course pages on the eLearn portal. Announcements will be made on the portal in a timely manner.
2. For Closed Book tests: No books or reference material of any kind will be permitted.
3. For Open Book exams: “open book” means text/ reference books (publisher copy only) and does not include any other learning material. No other learning material will be permitted during the open book examinations. For Detailed Guidelines refer to the attached document.

[EC3 Guidelines](#)

4. If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student should follow the procedure to apply for the Make-Up Test/Exam, which will be made available on the eLearn portal. The Make-Up Test/Exam will be conducted only at selected exam centres on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course handout, attend the online lectures, and take all the prescribed evaluation components such as Assignments/Quizzes, Mid-Semester Tests and Comprehensive Exams according to the evaluation scheme provided in the handout.





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